

NEXUSTRADE

AI-Powered Multi-Model Stock Trading Simulator

Executive Overview

NexusTrade is a full-stack, enterprise-grade stock trading simulator that harnesses multi-agent AI deliberation (Claude, GPT-4o, Gemini 1.5 Pro, DeepSeek Chat) to analyze stock market trends, deliberate across multiple rounds, and execute paper trades with intelligent Kelly-criterion position sizing. This document provides complete architectural, mathematical, and implementation details.

Project Metadata & Tech Stack

System Component	Technology & Details
Frontend Stack	Next.js 16 (App Router) · React 19 · TypeScript · Tailwind CSS v4 · Zustand
Backend Stack	FastAPI · Python 3.11 · SQLAlchemy 2 · SQLite / PostgreSQL · Pydantic v2
AI Intelligence	Claude Sonnet · GPT-4o · Gemini 1.5 Pro · DeepSeek Chat (2-Round Deliberation)
Market Data	Finnhub API (Real-time candles & news) with full deterministic mock fallback
Visualizations	Three.js particle canvas · TradingView lightweight-charts v5
Paper Wallet	\$100,000 starting virtual balance with FIFO cost basis tracking
Repository URL	https://github.com/sohangowda11/NexusTrade.git

1. Introduction to Stock Trading & Paper Trading

What Is a Stock?

A stock (equity share) represents a fractional ownership unit in a publicly traded corporation. When companies require growth capital, they issue shares via an Initial Public Offering (IPO). Owning shares entitles shareholders to proportionate equity ownership, voting rights, and potential dividend distributions.

How Prices Move: Market Microstructure

Stock prices fluctuate continuously based on real-time order matching between buyers (bids) and sellers (asks). Prices are driven by corporate earnings reports, macroeconomic metrics (interest rates, inflation, GDP growth), industry sector rotation, geopolitical events, and market sentiment.

Core Financial Terminology

Term	Technical Definition
Market Cap	Total market value of outstanding shares = Share Price × Shares Outstanding
Volatility	Standard deviation of price returns over a specified period. High volatility indicates greater price swings
Bid / Ask Spread	Bid = maximum price buyers offer. Ask = minimum price sellers accept. Spread = Ask – Bid
Volume	Total number of shares traded during a given timeframe. High volume validates price breakouts
P/E Ratio	Price-to-Earnings ratio = Share Price / Earnings Per Share (EPS). Evaluates valuation relative to earnings
Bull / Bear Market	Bull = sustained upward market trend (+20% from lows). Bear = sustained downward trend (-20% from highs)
Liquidity	The ease with which an asset can be converted into cash without affecting market price

Why Paper Trading Matters

Paper trading simulates live stock transactions using virtual funds while tracking real market prices. It allows traders and quantitative models to evaluate entry/exit rules, refine risk parameters, and measure strategy performance without risking capital. NexusTrade is a simulated trading environment — all trades use virtual currency.

2. Multi-Model AI Voting & Deliberation Engine

Architectural Overview

NexusTrade models stock market decision-making as a multi-agent consensus problem. Rather than depending on a single LLM, the engine queries four distinct state-of-the-art models in parallel, processes their independent outputs, and conducts a secondary discussion round where models evaluate peer perspectives.

Key Innovation: 2-Round Multi-Agent Deliberation

Single-agent LLM prompts suffer from hallucination and bias. NexusTrade introduces a structured 2-round protocol: Round 1 gathers un-anchored independent votes. Round 2 presents each model with an anonymized summary of peer reasoning, allowing models to refine or uphold their predictions based on cross-examination.

Model Input Prompt Structure

Every AI model receives a standardized contextual prompt containing:

- Ticker Symbol & Company Overview (e.g., AAPL, NVDA, TSLA)
- Real-time Price Metrics: Current price, day change %, 52-week high/low
- Technical Indicators: RSI (14), MACD Signal Line & Histogram, SMA 20, SMA 50, Bollinger Bands
- News Sentiment: Top 5 recent Finnhub market news headlines (last 48 hours)
- Strict Output Constraint: Structured JSON output contract

JSON Response Schema

```
{
  "direction": "up" | "down",
  "confidence": 0-100,
  "risk_level": "low" | "medium" | "high",
  "reasoning": "Concise justification"
}
```

Consensus Aggregation Algorithm

Signal Output	Mathematical Aggregation Rule
Final Direction	Majority vote among Round 2 predictions. Ties default to 'uncertain'
Final Confidence	Weighted average of confidence scores from models supporting the majority vote
Final Risk Level	Conservative Escalation: If ANY model flags 'high' risk, overall risk escalates to 'high'
Consensus Score	Percentage of models aligned with majority direction (e.g., 4/4 = 100%, 3/4 = 75%)

Deterministic Mock Mode Fallback

If API keys are omitted or network requests fail, the backend seamlessly routes to a deterministic mock service. Mock outputs are deterministically hashed from ticker symbol + model ID, providing consistent demonstration signals without requiring paid API credentials.

3. Paper Trading Engine & Risk Management

Virtual Portfolio & Account Balance

Every session initializes with \$100,000.00 in virtual USD cash balance, stored within the SQLite database. Users can perform simulated buy/sell orders, deposit/withdraw virtual capital, and track position performance in real-time.

FIFO Cost Basis & Average Price Calculation

When accumulating shares across multiple orders, the system updates the weighted average cost basis:

$$\text{New Average Cost} = (\text{Existing Shares} \times \text{Existing Avg Cost} + \text{New Shares} \times \text{New Price}) / \text{Total Shares}$$

Optimal Position Sizing: The Kelly Criterion

NexusTrade features automated position sizing based on Kelly Criterion gambling math. The Kelly fraction determines the optimal percentage of available cash to allocate:

$$\text{Kelly Fraction } f^* = \max(0, 2 \times p - 1) \text{ [where } p = \text{model_confidence} / 100]$$

Risk-Adjusted Position Caps

To prevent over-leveraging, the Kelly fraction is capped by the risk level of the analysis signal:

Signal Risk Level	Maximum Cash Allocation Cap	Example Execution (\$50,000 Cash, 70% Confidence → $f^* = 40\%$)
Low Risk	15% max allocation	$\min(40\%, 15\%) \times \$50,000 = \$7,500$ position size
Medium Risk	10% max allocation	$\min(40\%, 10\%) \times \$50,000 = \$5,000$ position size
High Risk	5% max allocation	$\min(40\%, 5\%) \times \$50,000 = \$2,500$ position size

4. Full-Stack System Architecture & Data Schemas

Frontend Technology Stack

Technology	Role & Implementation Purpose
Next.js 16	React Framework with App Router, SSR, and Client Component optimization
React 19	Modern UI library with concurrent rendering primitives
TypeScript	Strict static typing across API payloads, state stores, and component props
Tailwind CSS v4	CSS-first utility design system with customized fintech color tokens
Zustand	Lightweight persistent client state (wallet, holdings, trade history)
React Query	Server state management, background polling, and cache invalidation
Three.js	Canvas particle field hero visual background
lightweight-charts	TradingView high-performance canvas stock chart component

Backend Technology Stack

Framework / Library	Role & Implementation Purpose
FastAPI	Asynchronous Python web framework with OpenAPI validation
SQLAlchemy 2	Object-Relational Mapping (ORM) for database models
SQLite / PostgreSQL	Relational storage for trades, portfolios, and AI session history
scikit-learn	GradientBoostingClassifier for meta-model training on past AI predictions
httpx	Async HTTP client for parallel LLM and market API calls

Database Schema Definitions

Table Name	Primary Columns & Descriptions
Portfolio	id (PK), cash_balance (Float), created_at, updated_at
Holding	id (PK), portfolio_id (FK), ticker (String), shares (Float), avg_price (Float)
Trade	id (PK), portfolio_id (FK), ticker, action (BUY/SELL), shares, price, total, pnl, timestamp
AnalysisSession	id (PK), ticker, final_direction, final_confidence, final_risk, consensus_score, timestamp
AIVote	id (PK), session_id (FK), model_name, direction, confidence, risk_level, reasoning, round

5. Technical Analysis Indicators

RSI (Relative Strength Index - 14 Period)

RSI measures momentum on a 0-100 scale. $RSI > 70$ indicates overbought conditions (potential pullback), while $RSI < 30$ indicates oversold conditions (potential reversal bounce).

MACD (Moving Average Convergence Divergence)

MACD compares 12-period and 26-period EMAs against a 9-period Signal line. Positive histogram values confirm bullish momentum; negative values signal bearish pressure.

Bollinger Bands (20 Period, 2 Standard Deviations)

Bollinger Bands define high/low volatility envelopes around the 20-day SMA. Band squeezes indicate impending volatility expansion.

Simple Moving Averages (SMA 20 & SMA 50)

SMA 20 tracks short-term trends, while SMA 50 tracks medium-term direction. Golden Cross ($SMA\ 50 > SMA\ 200$) signifies long-term bullish bias.

6. Machine Learning Meta-Model Training

How Meta-Model Training Works

NexusTrade includes a scikit-learn meta-model (GradientBoostingClassifier) that learns which AI models are historically most accurate under specific market conditions. Rather than replacing LLMs, the meta-model acts as a data-driven referee.

Feature Matrix Representation

Feature Category	Input Features & Normalization
Model Directional Votes	claude_dir, gpt_dir, gemini_dir, deepseek_dir (Encoded: 1=UP, 0=DOWN, -1=NONE)
Model Confidence	claude_conf, gpt_conf, gemini_conf, deepseek_conf (Normalized 0.0 to 1.0)
Model Risk Flags	claude_risk, gpt_risk, gemini_risk, deepseek_risk (Encoded: 0=Low, 1=Med, 2=High)
Consensus & Time	consensus_score (0-100), weekday (0-6), hour_of_day (0-23)

Training Trigger & Execution

The meta-model requires a minimum of 20 historical sessions with verified 24-hour outcome labels before training. An automated background cron job (APScheduler) retrains the classifier every Sunday at 02:00 AM.

7. Local Setup & Execution Guide

Prerequisites

- Node.js 18+ & npm
- Python 3.11+
- Git

Step-by-Step Installation

Step	Action & Command
1. Clone Repository	git clone https://github.com/sohangowda11/NexusTrade.git
2. Setup Backend	cd NexusTrade/backend && pip install -r requirements.txt
3. Environment Config	copy backend\.env.example backend\.env
4. Launch Backend	cd backend && uvicorn app.main:app --reload --port 8000
5. Setup Frontend	cd frontend && npm install
6. Launch Frontend	cd frontend && npm run dev
7. Access App	Open http://localhost:3000 in your web browser

API Keys Configuration (Optional)

Provider	Environment Variable	Description
Finnhub	FINNHUB_API_KEY	Real-time stock candles & news headlines
Anthropic	ANTHROPIC_API_KEY	Claude Sonnet multi-agent voting
OpenAI	OPENAI_API_KEY	GPT-4o multi-agent voting
Google	GOOGLE_API_KEY	Gemini 1.5 Pro multi-agent voting
DeepSeek	DEEPSEEK_API_KEY	DeepSeek Chat multi-agent voting

8. Educational Disclaimers

IMPORTANT DISCLAIMER — NOT FINANCIAL ADVICE

NexusTrade is strictly an educational software simulation. Predictions, consensus signals, and Kelly criterion calculations generated by this software do NOT constitute financial advice, investment recommendations, or trading endorsements. Never make real-world financial investments based on AI model predictions or paper trading outputs.

NexusTrade Technical Documentation — Built with Next.js 16, FastAPI, and Multi-Agent LLMs.